

Calibration results

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Camera-system parameters:

cam0 (/camera_0/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.83227658 0.51043014 1014.00460977 1015.91873444 676.53169034

446.81372918] +- [0.00467633 0.02061217 0.13673649 0.13627592 0.11274725 0.12123621]

reprojection error: [-0.000001, 0.000000] +- [0.256012, 0.265840]

cam1 (/camera_1/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.93158406 0.40435005 1044.67214997 1046.03995762 682.29498249

447.03381061] +- [0.0048124 0.02068938 0.25735721 0.25647754 0.3274841 0.12032424]

reprojection error: [-0.000005, -0.000000] +- [0.251631, 0.264183]

cam2 (/camera_2/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.85309656 0.54800786 1023.49592412 1023.64866827 693.87828171

460.19385426] +- [0.00444159 0.02020206 0.12266133 0.12193332 0.16324517 0.14715503]

reprojection error: [0.000015, 0.000001] +- [0.261080, 0.285583]

cam3 (/camera_3/image_raw):

type: <class 'aslam_cv.libaslam_cv_python.DoubleSphereCameraGeometry'>

distortion: [] +- []

projection: [2.83306061 0.47079092 1013.04400699 1012.93396438 736.98469142

435.2130607] +- [0.00562834 0.02551101 0.2059652 0.20544251 0.12874117 0.17735451]

reprojection error: [0.000003, 0.000001] +- [0.245968, 0.271282]

baseline T_1_0:

q: [0.01090645 -0.70450242 -0.00222176 0.7096143] +- [0.00061957 0.00091182 0.00069047]

t: [0.04824305 0.00010445 -0.04886914] +- [0.00014616 0.00007455 0.00014496]

baseline T_2_1:

q: [-0.00446051 -0.70992267 0.00168049 0.70426351] +- [0.00138004 0.00246584 0.00130581]

t: [0.13136087 0.00049736 -0.13457055] +- [0.00031456 0.00038553 0.00055571]

baseline T_3_2:

q: [0.00127677 -0.70233654 -0.00437103 0.71183049] +- [0.00088261 0.0010595 0.00072127]

t: [0.04915733 0.00040846 -0.04822415] +- [0.0001714 0.0000947 0.00018837]

Target configuration

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Type: aprilgrid

Tags:

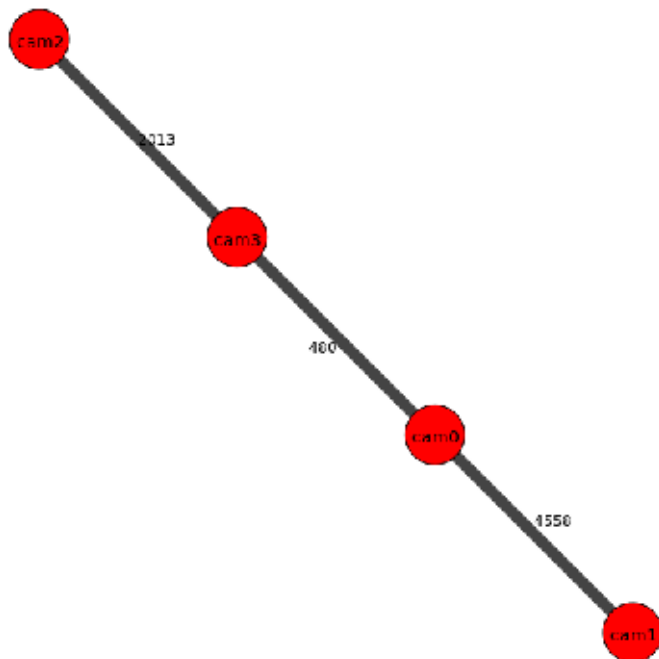
Rows: 6

Cols: 6

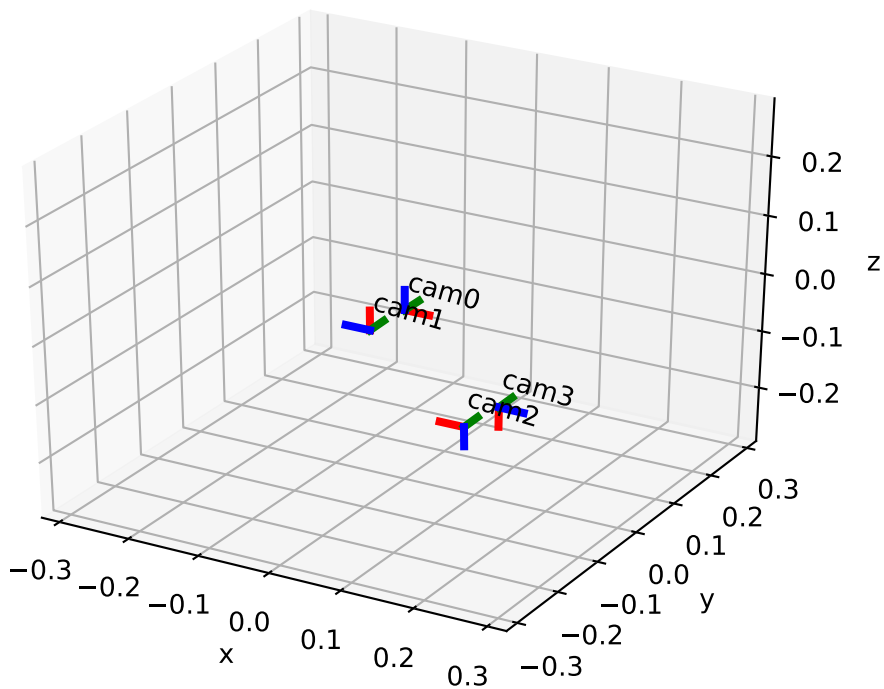
Size: 0.021 [m]

Spacing 0.0059997 [m]

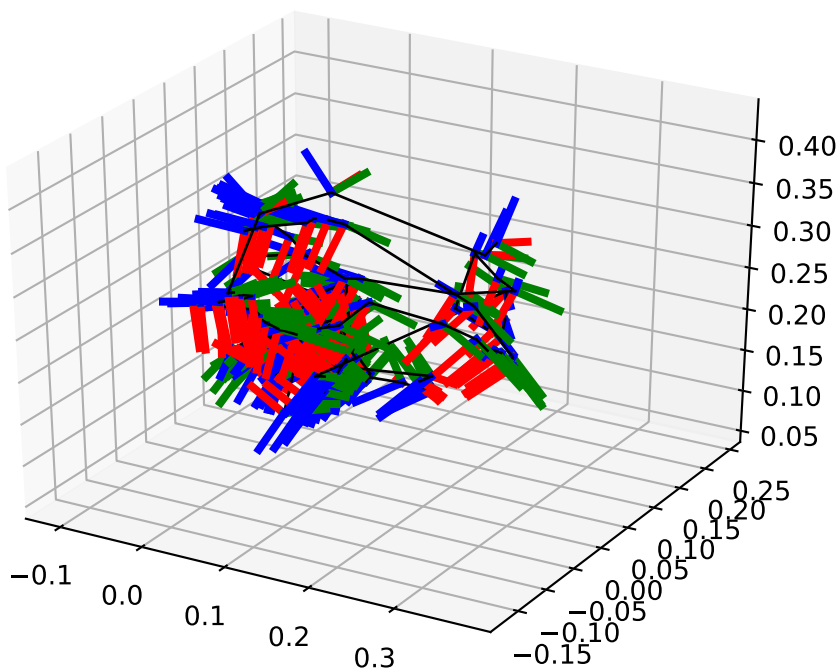
Inter-camera observations graph (edge weight=#mutual obs.)



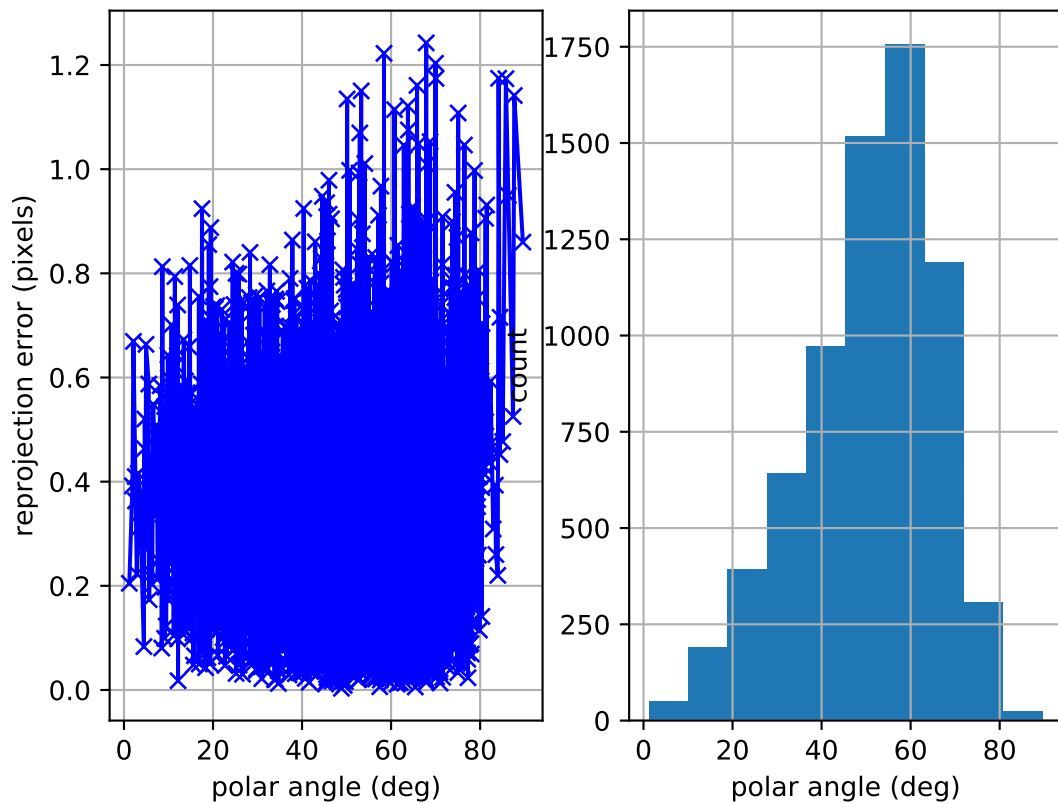
camera system



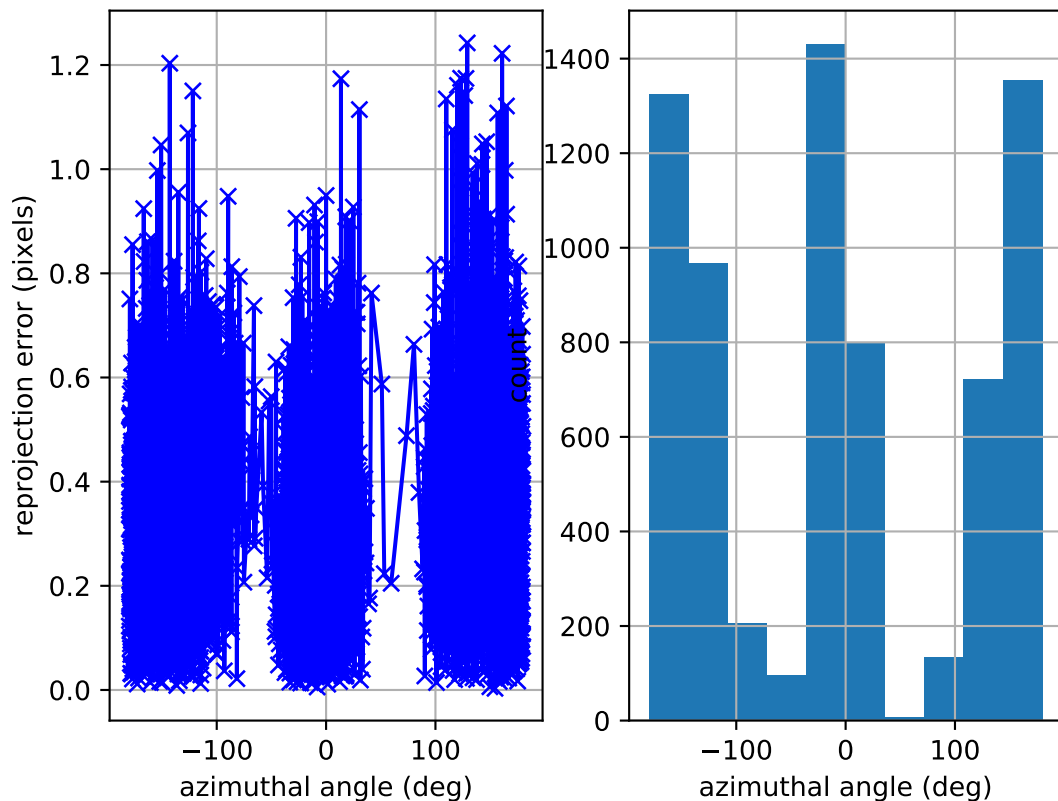
cam0: estimated poses



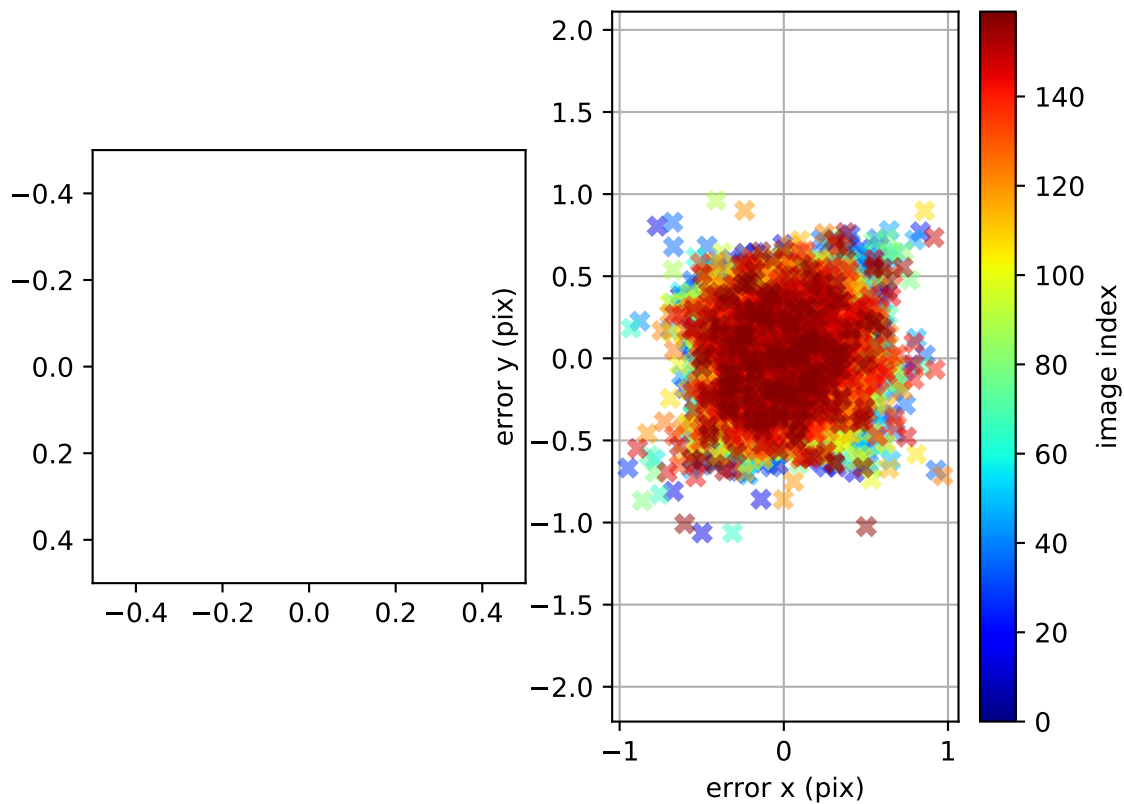
cam0: polar error



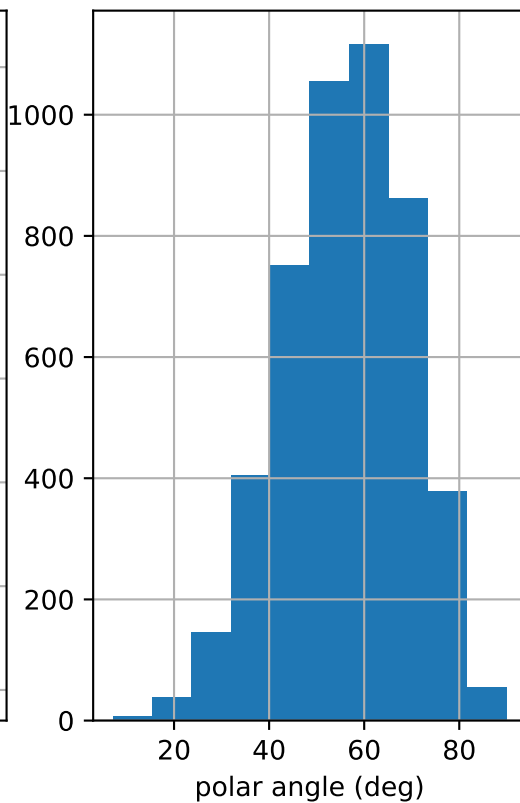
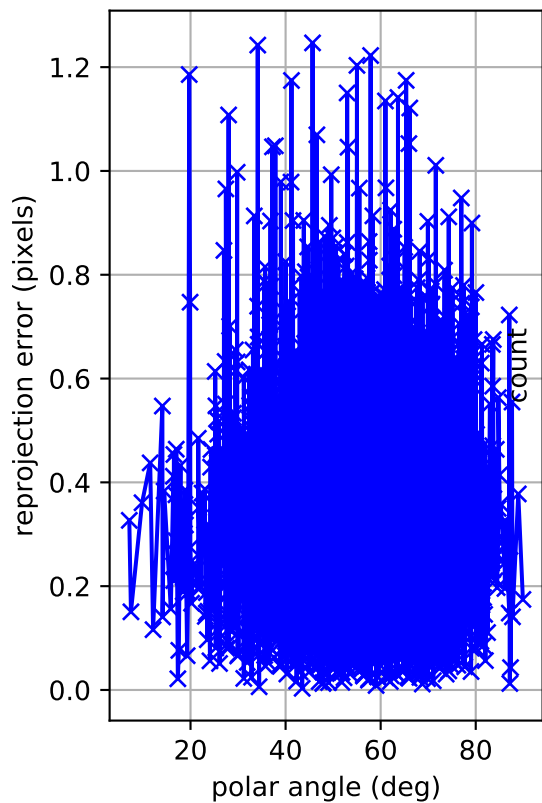
cam0: azimuthal error



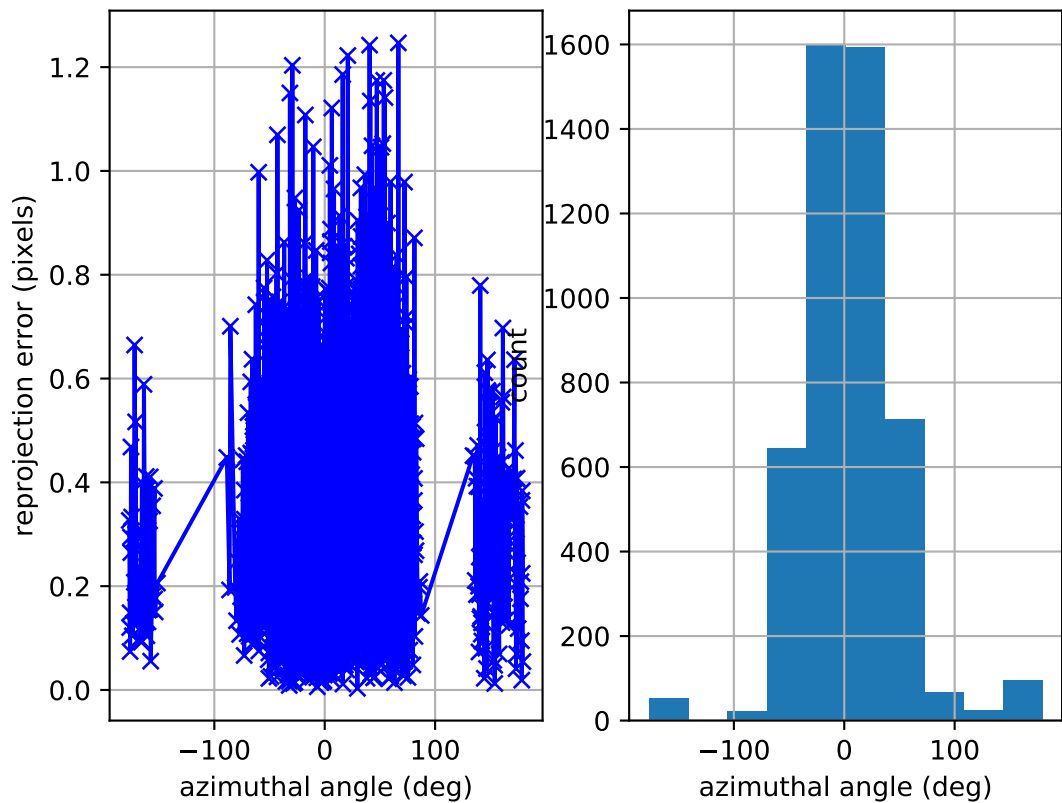
cam0: reprojection errors



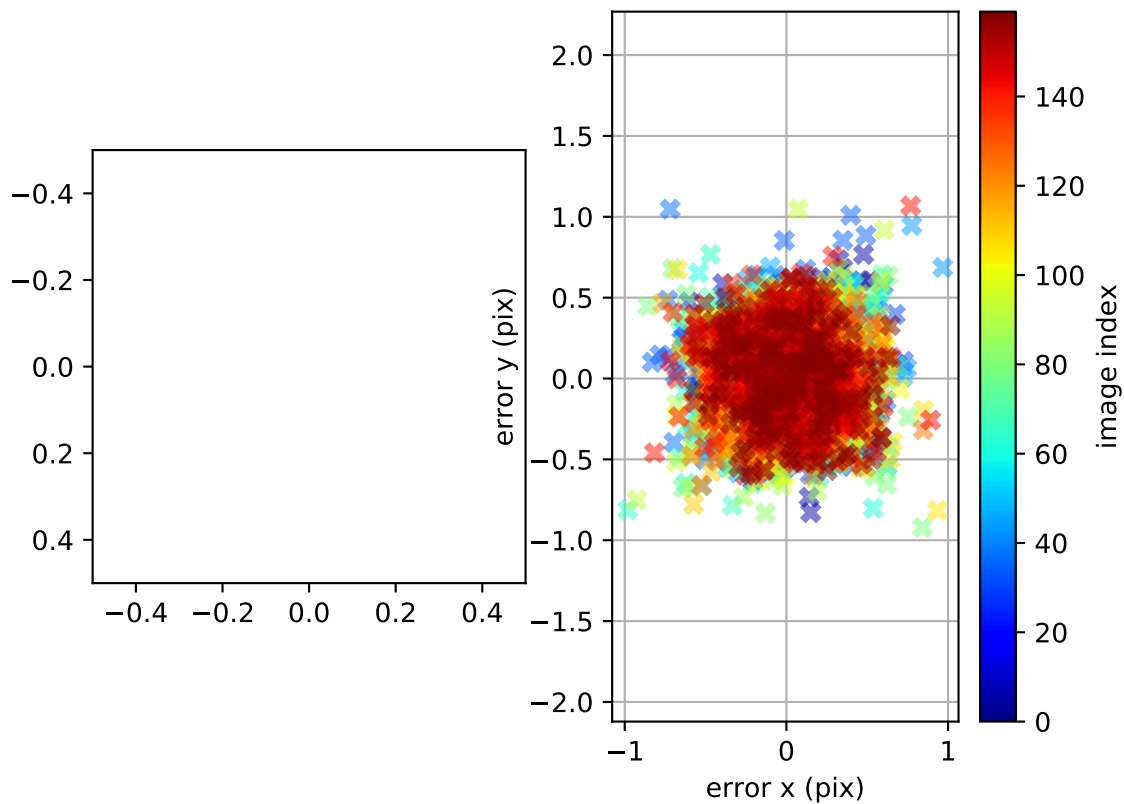
cam1: polar error



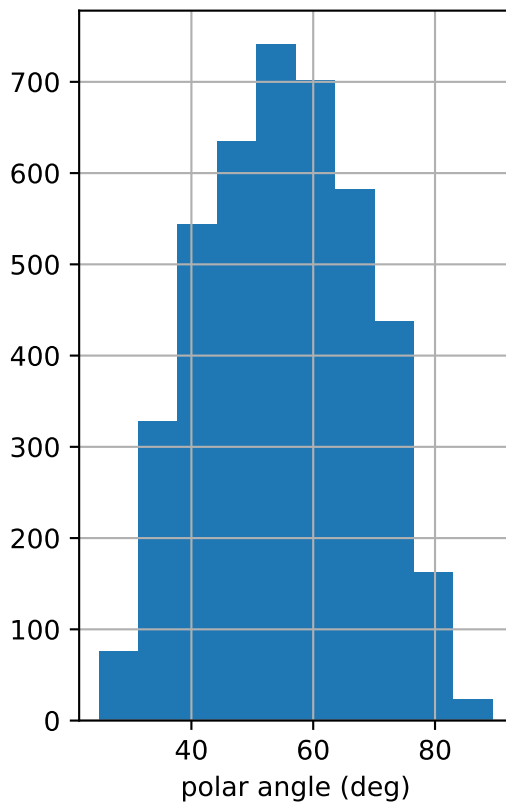
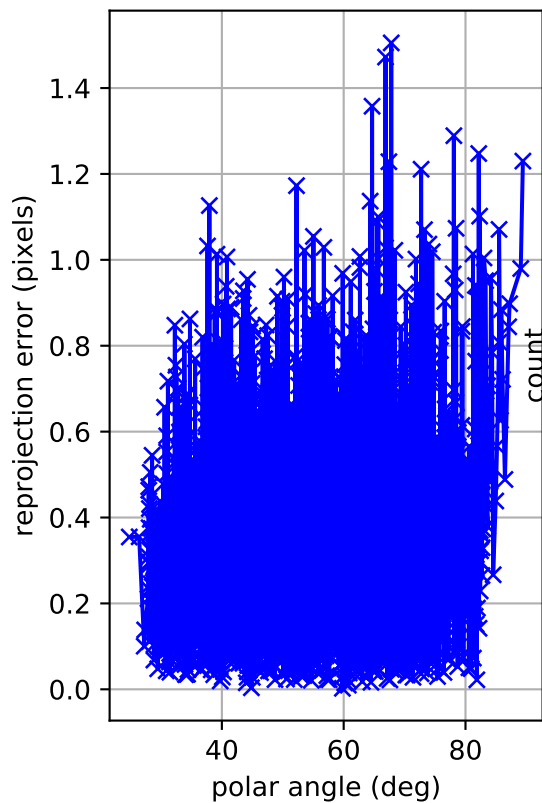
cam1: azimuthal error



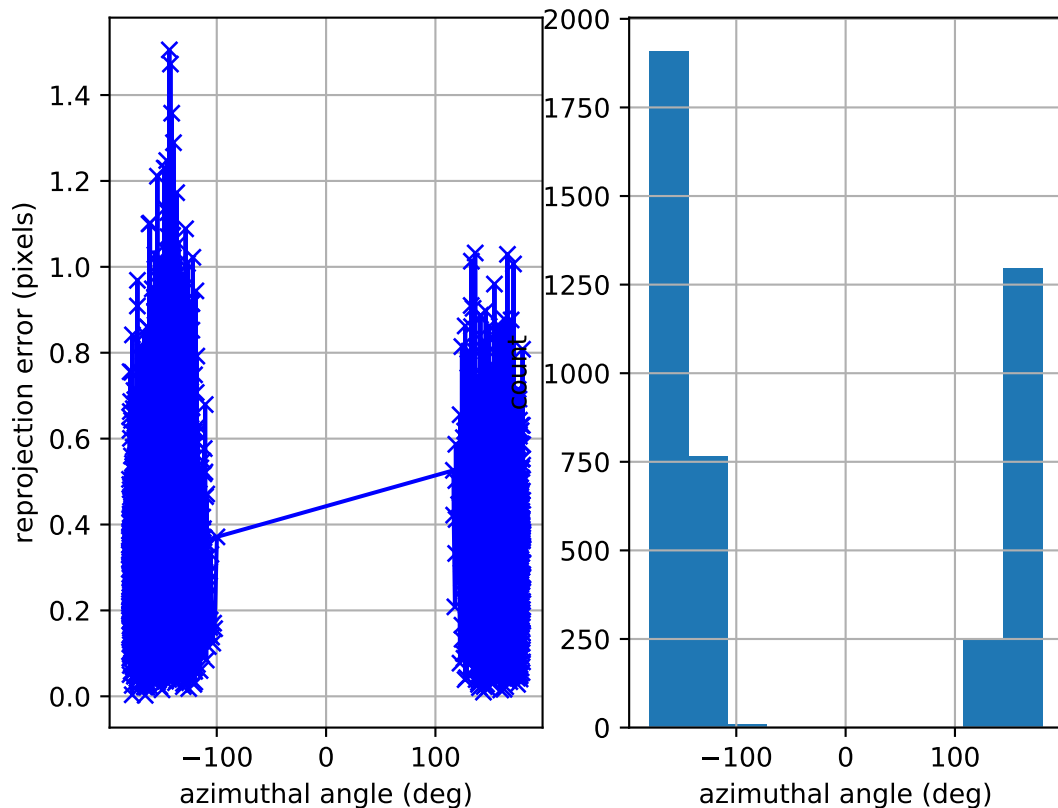
cam1: reprojection errors



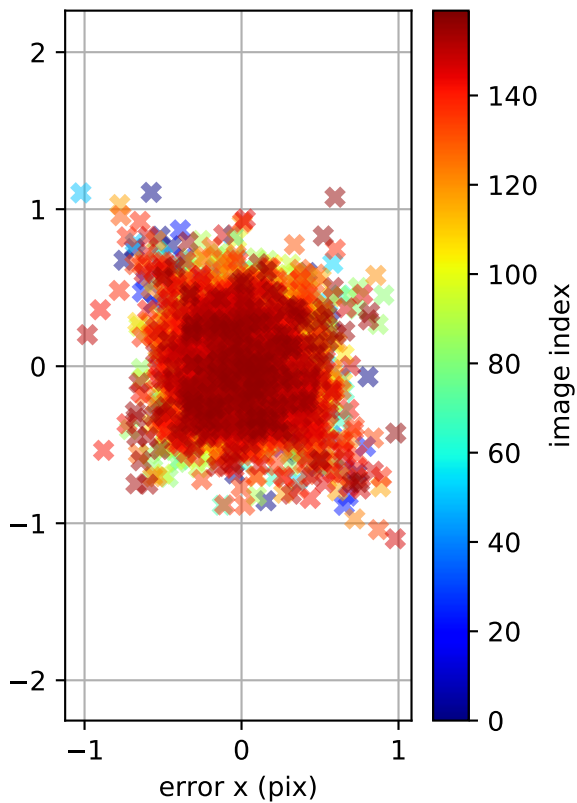
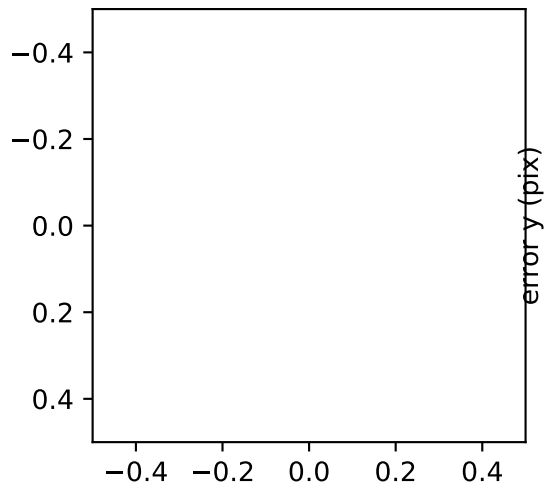
cam2: polar error



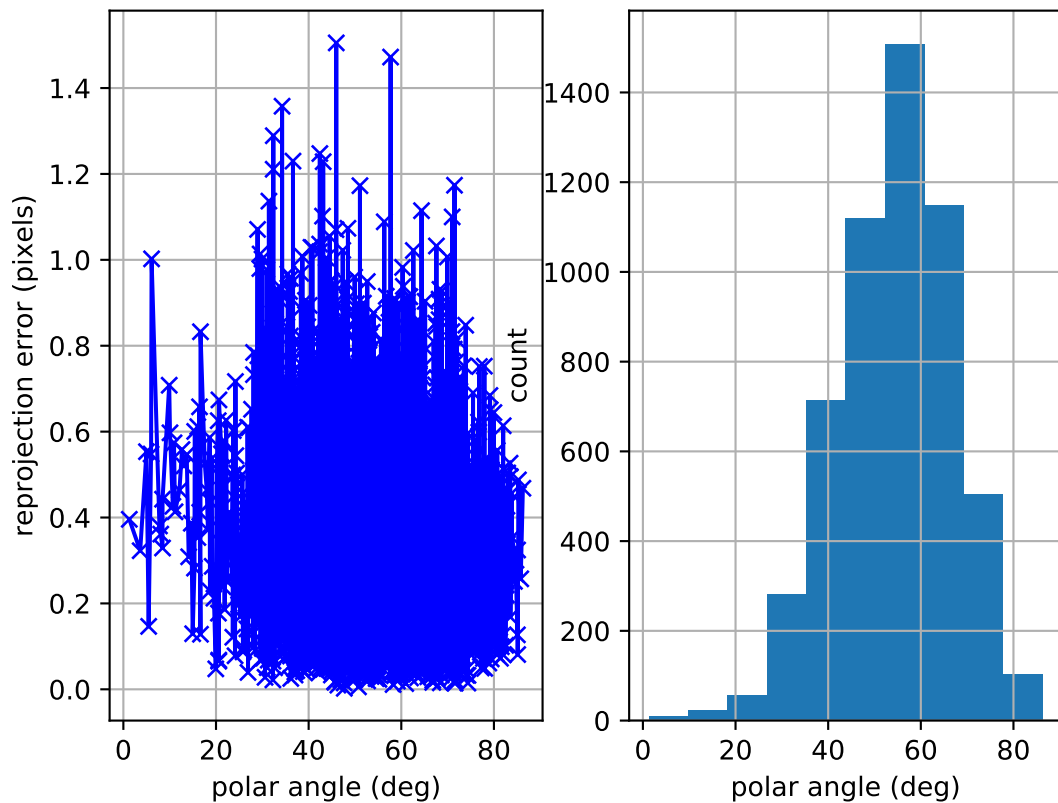
cam2: azimuthal error



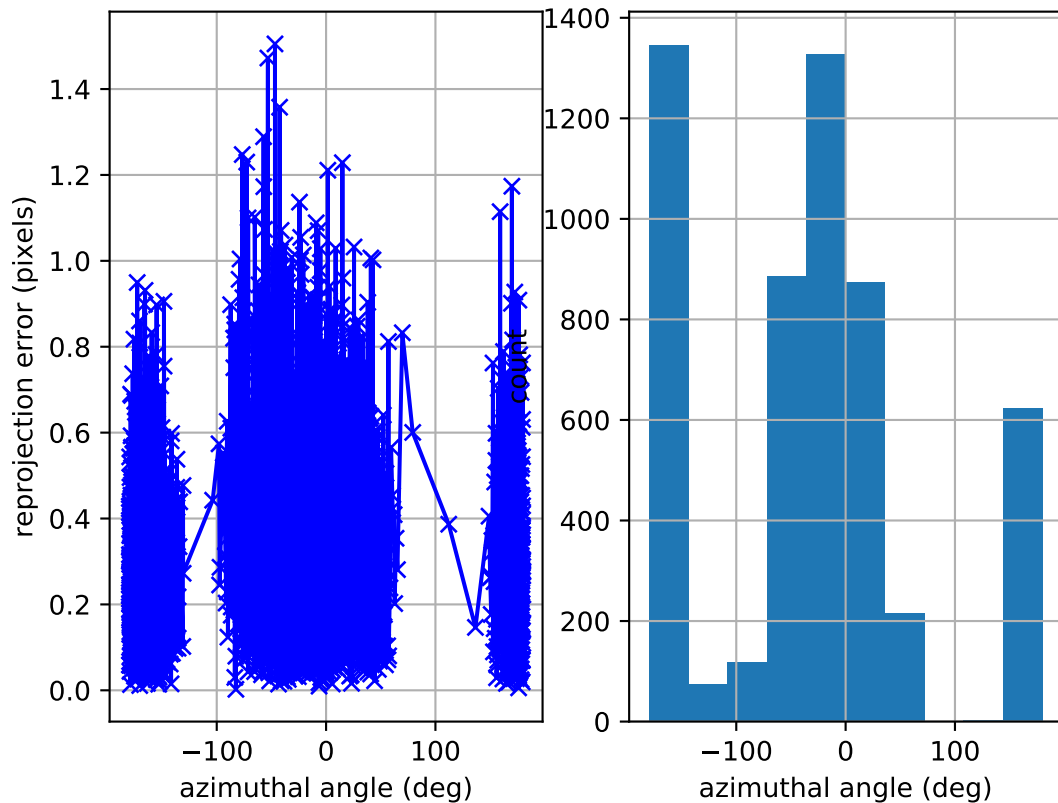
cam2: reprojection errors



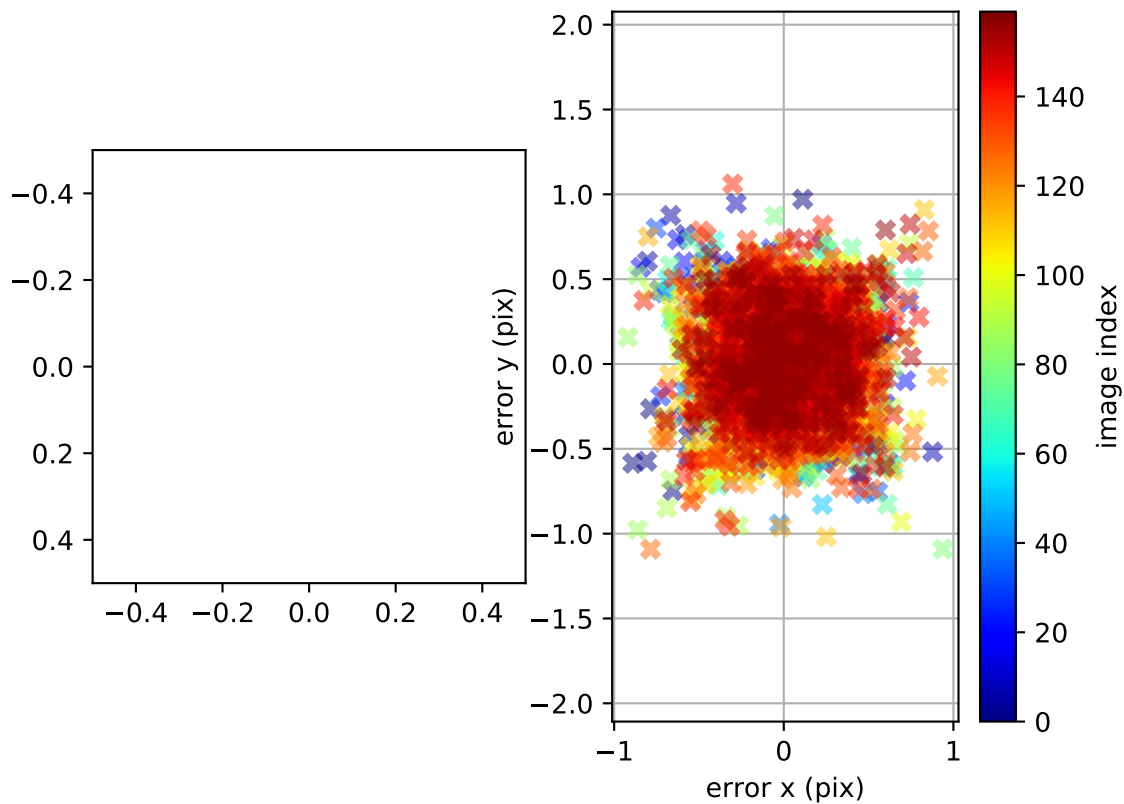
cam3: polar error



cam3: azimuthal error

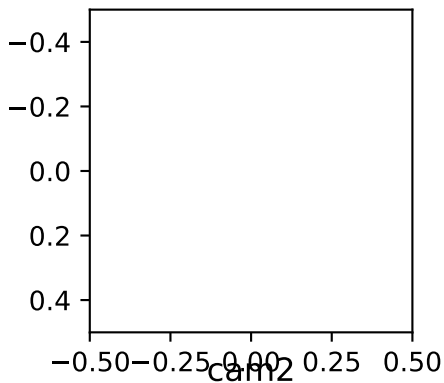


cam3: reprojection errors

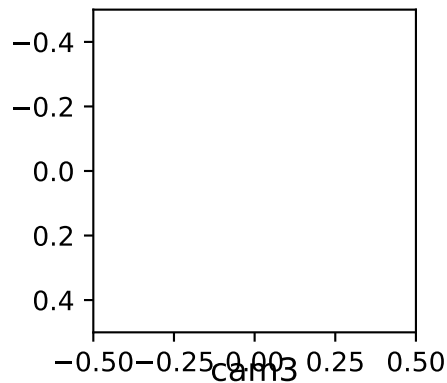


Location of removed outlier corners

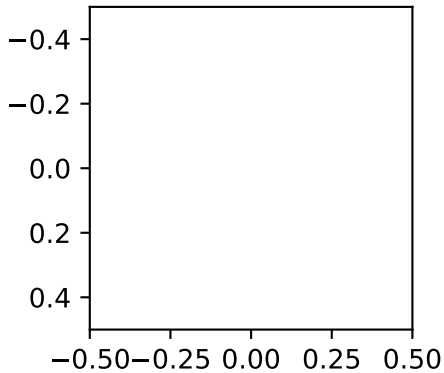
cam0



cam1



cam2



cam3

